

SYEDA NEHA ZAFAR

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SUMMARY

AI Engineer specializing in Generative AI, RAG, and applied NLP/CV systems, with hands-on production exposure spanning conversational AI, PII detection and deployment, OCR pipelines, semantic search, and voice-biometrics architecture. Comfortable owning a problem end-to-end: research and literature review, model selection and fine-tuning, evaluation, and deployment to real users (Hugging Face Spaces, containerized services, cloud) across AWS, Azure, and Google Cloud.

TECHNICAL SKILLS

AI: LLMs, RAG Pipelines, Parent Document Retrieval, Autonomous Agents & Agent Security Auditing, LangChain, Prompt Engineering, Vector Databases, Embeddings, Semantic Search, GLiNER, DPO, PyTorch, TensorFlow, Scikit-learn, PyCaret, Neural Networks (CNN, ResNet-50), Sentence-Transformers, Model Evaluation, Feature Engineering, Vertex AI, PaddleOCR, OpenCV, Roboflow Annotation, Object Detection

Cloud & Deployment: Docker, Kubernetes (GKE), AWS, Microsoft Azure, Google Cloud Platform, Compute Engine, Hugging Face Spaces, FastAPI, Streamlit

Programming & Data: Python, SQL (MySQL, PostgreSQL), Node.js, Express, Django, MongoDB, Java, C#, Git/GitHub, REST APIs, Linux

WORK EXPERIENCE

AI Engineer — Proxima AI

June 2026 – Present

- Designed and deployed a conversational AI chatbot featuring context-aware dynamic state management, robust intent recognition, and automated fallback logic for real-time user interaction.
- Fine-tuned and deployed a zero-shot Urdu PII redaction model using GLiNER architecture for regional sensitive data formats; published a live demo on Hugging Face Spaces.
- Built a candidate-to-job matching engine utilizing NLP entity extraction and high-dimensional vector embeddings to automate resume screening via contextual fit scoring.
- Engineered an automated document-processing pipeline integrating OpenCV image pre-processing (deskewing, adaptive thresholding) and PaddleOCR to extract tabular receipt data into structured JSON.
- Conducted security audits on autonomous LLM agents, identifying and mitigating context injection, prompt exploitation, and data leakage risks across vector databases and third-party APIs.

Data Analyst Intern — Elevvo

Aug 2025 – Sep 2025

- Executed EDA pipelines in Python (Pandas, Seaborn) for data cleaning, anomaly detection, and statistical visualization.
- Built RFM (Recency, Frequency, Monetary) analytics models on retail datasets to identify behavioral segments and inform retention strategy.
- Wrote SQL queries with JOINS, aggregations, and GROUP BY logic to analyze revenue across business units.
- Designed interactive Power BI dashboards with slicers and metric targets for operational reporting.

Software Developer Intern — JS Bank

Apr 2025 – Jun 2025

- Built a real-time NLP lip-synchronization module using DeepSeek multi-modal APIs (TTS/STT) linked to automated avatar movement.
- Developed RESTful APIs (Node.js, Express, MongoDB) with CRUD operations, request logging, and input validation.
- Configured containerized deployments and Python automation scripts delivered via secure SFTP/SSH and jump-server paths.

SELECTED PROJECTS

Makkhi Meter 2.0 — Final Year Project

Mar 2024 – Jan 2025

- Built a distributed ML-based vision system using custom CNN, SVM, and KNN models for automated phenotypic classification.

Pneumonia Diagnostic Platform — Personal Project

6 Weeks

- Developed an AI radiography analysis pipeline using a ResNet-50 architecture for multi-class classification across 2,500 images.

EDUCATION

Bachelor of Software Engineering — Usman Institute of Technology (NED Affiliated) (2021 – 2025)

CERTIFICATIONS

- Kaggle: Feature Engineering, Intro to Deep Learning, Computer Vision, Intro to AI Ethics, Geospatial Analysis
- Microsoft: Azure for AI and Machine Learning
- Meta: Front-End Developer, Advanced React
- Google: Introduction to AI Agents, Prompt Design in Agent Platform, Build a Website on Google Cloud, Foundations of Project Management, Agile Project Management